```
> restart; #the thermal's flux ; 8 sec
```

```
> eq1 := diff(p(r,t),t$2) + V·diff((p(r,t)·V),r$2) - (V^2·p(r,t))/R = 0;
```

$$eq1 := \frac{\partial^2}{\partial t^2} p(r,t) + V^2 \left(\frac{\partial^2}{\partial r^2} p(r,t) \right) - \frac{V^2 p(r,t)}{R} = 0 \tag{1}$$

```
> eq2 := diff(p(r,t),t) + diff(p(r,t)·V, r) = 0;
```

$$eq2 := \frac{\partial}{\partial t} p(r,t) + \left(\frac{\partial}{\partial r} p(r,t) \right) V = 0 \tag{2}$$

```
> iv1 := p(2·R, 0) = 0, D[1](p)(0,0) = -R;
```

$$iv1 := p(2\,R, 0) = 0, \mathrm{D}_1(p)(0,0) = -R \tag{3}$$

```
> Sol := pdsolve([eq1, eq2, iv1]);
```

$$Sol := p(r,t) = \frac{\sqrt{2}\, R^{3/2} \left(e^{\frac{(Vt + 2R - r)\sqrt{2}}{\sqrt{R}}} - 1 \right) e^{-\frac{\sqrt{2}\,(Vt - r)}{2\sqrt{R}}}}{e^{2\sqrt{2}\sqrt{R}} + 1} \tag{4}$$

```
> Sol
```

$$p(r,t) = \frac{\sqrt{2}\, R^{3/2} \left(e^{\frac{(Vt + 2R - r)\sqrt{2}}{\sqrt{R}}} - 1 \right) e^{-\frac{\sqrt{2}\,(Vt - r)}{2\sqrt{R}}}}{e^{2\sqrt{2}\sqrt{R}} + 1} \tag{5}$$

```
> change_of_vars := { r = rb·ρ }
```

$$change_of_vars := \{ r = rb\,\rho \} \tag{6}$$

```
> tmp := PDEtools:-dchange(change_of_vars, [eq1, eq2, iv1], [ρ], params = rb);
```

$$tmp := \left[\frac{\partial^2}{\partial t^2} p(\rho,t) + \frac{V^2 \left(\frac{\partial^2}{\partial \rho^2} p(\rho,t) \right)}{rb^2} - \frac{V^2 p(\rho,t)}{R} = 0, \frac{\partial}{\partial t} p(\rho,t) + \frac{\left(\frac{\partial}{\partial \rho} p(\rho,t) \right) V}{rb} = 0, p(2\,R, 0) = 0, \mathrm{D}_1(p)(0,0) = -R \right] \tag{7}$$

```
> eq3 := diff(p(ρ,t),t$2) + (V^2·diff(p(ρ,t),ρ$2))/rb^2 - (V^2·p(ρ,t))/R = 0;
```

$$eq3 := \frac{\partial^2}{\partial t^2} p(\rho,t) + \frac{V^2 \left(\frac{\partial^2}{\partial \rho^2} p(\rho,t) \right)}{rb^2} - \frac{V^2 p(\rho,t)}{R} = 0 \tag{8}$$

```
> eq4 := diff(p(ρ,t),t) + (diff(p(ρ,t),ρ)·V)/rb = 0, p(2·R, 0) = 0;
```

$$eq4 := \frac{\partial}{\partial t} p(\rho, t) + \frac{\left(\frac{\partial}{\partial \rho} p(\rho, t)\right) V}{rb} = 0, p(2R, 0) = 0 \tag{9}$$

```
> Sol := pdsolve([eq3, eq4, iv1]);
```

$$Sol := p(\rho, t) = \frac{\sqrt{2}\, R^{3/2} \left(e^{\frac{2\left(\left(R - \frac{\rho}{2}\right) rb + \frac{Vt}{2}\right)\sqrt{2}}{\sqrt{R}}} - 1 \right) e^{-\frac{\sqrt{2}\,(Vt - rb\,\rho)}{2\sqrt{R}}}}{\left(e^{2\sqrt{2}\,\sqrt{R}\,rb} + 1\right) rb} \tag{10}$$

```
> solve({ sqrt(2)*R^(3/2)*(exp(2*(sqrt(2)*((R-rho/2)*rb+V*t/2))/sqrt(R))-1)*exp(-sqrt(2)*(V*t-rb*rho)/(2*sqrt(R)))/((exp(2*rb*sqrt(2)*sqrt(R))+1)*rb) = 1, D[2](p)(1,0) = V, rho = 1 });
```

$$\left\{ R = R,\ V = D_2(p)(1,0),\ rb = rb,\ \rho = 1,\ t = -\frac{1}{D_2(p)(1,0)} \left(\ln\left(RootOf\left(2\,_Z^2\, R^{3/2} + \left(\sqrt{2}\, rb\, e^{2\sqrt{2}\,\sqrt{R}\,rb} + \sqrt{2}\, rb \right) _Z - 2\, R^{3/2}\, e^{2\sqrt{2}\,\sqrt{R}\,rb} \right) \right) \sqrt{2}\,\sqrt{R} - rb \right) \right\} \tag{11}$$

```
> restart;
```

```
> eq1 := (2*m)/h^2*E*f(r,t) + R/r*diff((f(r,t)), r$2) - 1/(a^2*c^2)*diff(f(r,t), t$2) = 0;
```

$$eq1 := \frac{2\, m\, E f(r,t)}{h^2} + \frac{R\left(\frac{\partial^2}{\partial r^2} f(r,t)\right)}{r} - \frac{\frac{\partial^2}{\partial t^2} f(r,t)}{a^2\, c^2} = 0 \tag{12}$$

```
> iv1 := f(2*R, t) = 0, D[1](f)(0,0) = R, D[2](g)(rb, 0) = a*c;
```

$$iv1 := f(2R, t) = 0,\ D_1(f)(0,0) = R,\ D_2(g)(rb, 0) = a\, c \tag{13}$$

```
> change_of_vars := { r = rb*rho, f(r,t) = g(rho,t)*(2*R - rb*rho)/(2*R) };
```

$$change_of_vars := \left\{ r = rb\,\rho,\ f(r,t) = \frac{g(\rho,t)\,(-rb\,\rho + 2R)}{2R} \right\} \tag{14}$$

```
> tmp := PDEtools:-dchange(change_of_vars, [eq1, iv1], [g(rho,t), rho]);
```

$$tmp := \left[\frac{m\, E\, g(\rho,t)\,(-rb\,\rho + 2R)}{h^2\, R} \right] \tag{15}$$

$$+\frac{R\left(\frac{\left(\frac{\partial^2}{\partial \rho^2}\,g(\rho,t)\right)(-rb\,\rho+2R)}{2R}-\frac{\left(\frac{\partial}{\partial \rho}\,g(\rho,t)\right)rb}{R}\right)}{rb^3\,\rho}$$

$$-\frac{\left(\frac{\partial^2}{\partial t^2}\,g(\rho,t)\right)(-rb\,\rho+2R)}{2a^2c^2R}=0,\ 0=0,\ \frac{D_1(g)(0,0)-\frac{g(0,0)\,rb}{2R}}{rb}=R,\ D_2(g)(rb,$$

$$0)=a\,c$$

> $eq2 := tmp[1];$

$$eq2 := \frac{m\,E\,g(\rho,t)\,(-rb\,\rho+2R)}{h^2R}$$

$$+\frac{R\left(\frac{\left(\frac{\partial^2}{\partial \rho^2}\,g(\rho,t)\right)(-rb\,\rho+2R)}{2R}-\frac{\left(\frac{\partial}{\partial \rho}\,g(\rho,t)\right)rb}{R}\right)}{rb^3\,\rho}$$

$$-\frac{\left(\frac{\partial^2}{\partial t^2}\,g(\rho,t)\right)(-rb\,\rho+2R)}{2a^2c^2R}=0 \tag{16}$$

> $eq3 := tmp[3];$

$$eq3 := \frac{D_1(g)(0,0)-\frac{g(0,0)\,rb}{2R}}{rb}=R \tag{17}$$

> $eq4 := tmp[4];$

$$eq4 := D_2(g)(rb,0)=a\,c \tag{18}$$

> $\#eq5 := tmp[5];$

> $iv2 := \frac{D_1(g)(0,0)-\frac{g(0,0)\,rb}{2R}}{rb}=R,\ D_2(g)(rb,0)=a\,c;$

$$iv2 := \frac{D_1(g)(0,0)-\frac{g(0,0)\,rb}{2R}}{rb}=R,\ D_2(g)(rb,0)=a\,c \tag{19}$$

> $sys := [eq2, iv2]:\ Sol := pdsolve(sys, HINT=`*`);$

(20)

$$Sol := g(\rho, t) = -\left(\text{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} \rho \right) \left(\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} \Gamma\left(\frac{2}{3}\right) a \left(e^{\sqrt{-_c_2}\,t} - e^{-\sqrt{-_c_2}\,t} \right) \left(-\frac{rb^2}{2} + R \right) c\, RootOf\left(_Z^2 - _c_2\right) - \frac{1}{3}\left(2\,_c_2\,3^{1/3} R^2 \pi\, rb \left(e^{\sqrt{-_c_2}\,t} + e^{-\sqrt{-_c_2}\,t} \right) \text{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} rb \right) \right) \right) \Big/ \left(\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} (rb\,\rho - 2\,R)\,_c_2\,\Gamma\left(\frac{2}{3}\right) \text{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} rb \right) \right) \tag{20}$$

$$> g(r, t) := -\left(\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} \Gamma\left(\frac{2}{3}\right) \left(-\frac{rb^2}{2} + R \right) \cdot \sqrt{-_c_2} \cdot a\,c \left(e^{\sqrt{-_c_2}\,t} - e^{-\sqrt{-_c_2}\,t} \right) - \frac{1}{3}\left(2\,_c_2\,3^{1/3} R^2 \pi\, rb \left(e^{\sqrt{-_c_2}\,t} + e^{-\sqrt{-_c_2}\,t} \right) \text{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2 - _c_2\,h^2\right) rb^3}{R\,a^2\,c^2\,h^2} \right)^{1/3} rb \right) \right) \right) \text{AiryBi}\left(
$$

$$-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\Bigg)\Bigg)\Bigg/\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}(rb\,\rho$$

$$-2\,R)\,_c_2\,\Gamma\left(\frac{2}{3}\right)\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\Bigg);$$

$$g:=(r,t)\mapsto -\Bigg(\Bigg(\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\Gamma\left(\frac{2}{3}\right)\left(-\frac{rb^2}{2}+R\right)\sqrt{_c_2}\,a\,c\left(e^{\sqrt{_c_2}\,t}\right. \tag{21}$$

$$\left.-e^{-\sqrt{_c_2}\,t}\right)$$

$$-\frac{1}{3}\Bigg(2\,_c_2\,3^{1/3}R^2\,\pi\,rb\left(e^{\sqrt{_c_2}\,t}+e^{-\sqrt{_c_2}\,t}\right)\mathrm{AiryBi}\Bigg($$

$$-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\Bigg)\Bigg)\mathrm{AiryBi}\Bigg($$

$$-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\Bigg)\Bigg)\Bigg/\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}(rb\,\rho$$

$$-2\,R)\,_c_2\,\Gamma\left(\frac{2}{3}\right)\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-_c_2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\Bigg)$$

> $G:=subs\left(\sqrt{_c_2}=C1,\ _c_2=C1^2,\ \Gamma\left(\frac{2}{3}\right)=1.354117939,\ g(\rho,t)\right);$

$$G:=-\Bigg(0.7384881119\Bigg(1.354117939\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(-\frac{rb^2}{2}\right. \tag{22}$$

$$\left.+R\right)C1\,a\,c\left(e^{C1\,t}-e^{-C1\,t}\right)$$

$$-\frac{2\,C1^2\,3^{1/3}R^2\,\pi\,rb\left(e^{C1\,t}+e^{-C1\,t}\right)\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)}{3}\Bigg)$$

$$\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\right)\Bigg/\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(rb\,\rho-2\,R\right)\,C1^2\,\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\right)$$

$$=$$

```
> G1 := subs(Γ(2/3) = 1.354117939, -(((((2*E*m*a^2*c^2 - C1^2*h^2)*rb^3)/(R*a^2*c^2*h^2))^(1/3)*Γ(2/3)*(-(rb^2)/2 + R)*C1*a*c*(e^(C1*t) - e^(-C1*t)) - (2*C1^2*3^(1/3)*R^2*π*rb*(e^(C1*t) + e^(-C1*t))*AiryBi(-(((2*E*m*a^2*c^2 - C1^2*h^2)*rb^3)/(R*a^2*c^2*h^2))^(1/3)*rb))/3)*AiryBi(-(((2*E*m*a^2*c^2 - C1^2*h^2)*rb^3)/(R*a^2*c^2*h^2))^(1/3)*ρ))/(((((2*E*m*a^2*c^2 - C1^2*h^2)*rb^3)/(R*a^2*c^2*h^2))^(1/3)*(rb*ρ - 2*R)*C1^2*Γ(2/3)*AiryBi(-(((2*E*m*a^2*c^2 - C1^2*h^2)*rb^3)/(R*a^2*c^2*h^2))^(1/3)*rb))));
```

$$G1 := -\left(0.7384881119\left(1.354117939\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(-\frac{rb^2}{2}+R\right)\,C1\,a\,c\left(e^{C1\,t}-e^{-C1\,t}\right)-\frac{2\,C1^2\,3^{1/3}\,R^2\,\pi\,rb\left(e^{C1\,t}+e^{-C1\,t}\right)\,\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)}{3}\right)\right) \tag{23}$$

$$\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\rho\right)\Bigg/\left(\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}\left(rb\,\rho-2\,R\right)\,C1^2\,\mathrm{AiryBi}\left(-\left(\frac{\left(2\,E\,m\,a^2\,c^2-C1^2\,h^2\right)\,rb^3}{R\,a^2\,c^2\,h^2}\right)^{1/3}rb\right)\right)$$

> $m := 9.109383560\,10^{-31}$; $\epsilon := 8.854187815\,10^{-12}$; $h := \frac{6.626070040\,10^{-34}}{2\cdot\mathrm{Pi}}$; $e := 1.602176620\,10^{-19}$; $c := 299792458$; $rb := 5.291772105\,10^{-11}$; $Tq := 1.76536\cdot10^{-19}$; $E := 2.179\cdot10^{-18}$; $a := 0.007297352566$; $R := 3.9228\cdot10^{22}$; $ms := -9.284764612\,10^{-24}$; $hm := \frac{3.151228184\,10^{-27}}{2\cdot\mathrm{Pi}}$; $Sq := 4.3\cdot10^{-19}$; $C1 := 1$;

$$m := 9.109383560\,10^{-31}$$
$$\epsilon := 8.854187815\,10^{-12}$$
$$h := 1.054571800\,10^{-34}$$
$$e := 1.602176620\,10^{-19}$$
$$c := 299792458$$
$$rb := 5.291772105\,10^{-11}$$
$$Tq := 1.765360000\,10^{-19}$$
$$E := 2.179000000\,10^{-18}$$
$$a := 0.007297352566$$
$$R := 3.566000000\,10^{22}$$
$$ms := -9.284764612\,10^{-24}$$
$$hm := 5.015335422\,10^{-28}$$
$$Sq := 4.300000000\,10^{-19}$$
$$C1 := 1$$

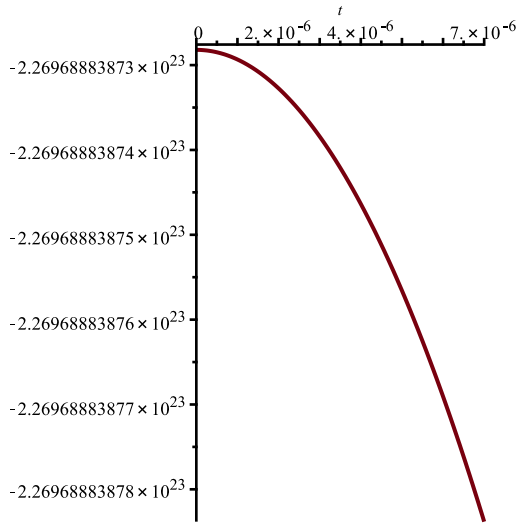
(24)

> $\#\rho := 10^{-19}$; $evalf\left(\Gamma\left(\frac{2}{3}\right)\right)$;$\#\rho\ constant$;

$$\rho := \frac{1}{10000000000000000000}$$
$$1.354117939$$

(25)

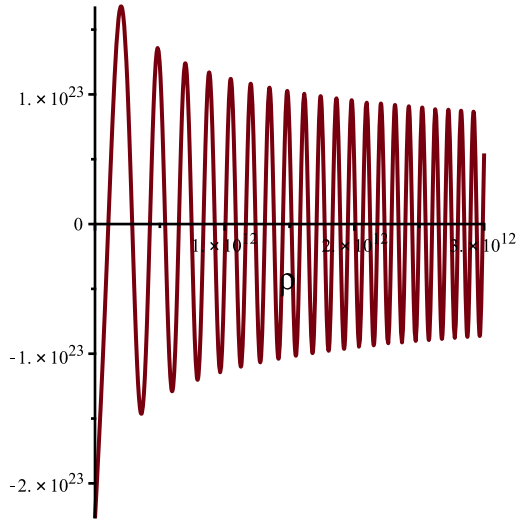
> $plot\left(G1, t = 0..7\cdot10^{-6}\right)$;



> $t := 0$;#*t constant; Wave function*

$$t := 0 \tag{26}$$

> $plot(G, \rho = 0..3 \cdot 10^{12})$;



>